

Ectoparasitism and stress hormones: strategy of host exploitation, common host–parasite history and energetics matter

Justin R. St. Juliana^{1,2,3*}, Irina S. Khokhlova⁴, Nadja Wielebnowski^{5,6}, Burt P. Kotler² and Boris R. Krasnov²

¹Department of Biology, Indiana State University, 200 N. 7th St., Terre Haute, IN 47809, USA; ²Mitrani Department of Desert Ecology, Swiss Institute for Dryland Environmental and Energy Research, Jacob Blaustein Institutes for Desert Research, Ben-Gurion University of the Negev, Midreshet Ben-Gurion, Israel; ³Department of Life and Physical Science, Ivy Tech Community College Wabash Valley, 8000 S. Education Drive, Terre Haute, IN 47802, USA; ⁴Wyller Department of Dryland Agriculture, French Associates Institute for Agriculture and Biotechnology of Drylands Jacob Blaustein Institutes for Desert Research, Ben-Gurion University of the Negev, Midreshet Ben-Gurion, Israel;

⁵Department of Conservation Science, Chicago Zoological Society/Brookfield Zoo, 3300 Golf Road, Brookfield, IL 60513, USA; and ⁶Conservation Research Division, Oregon Zoo, 4001 SW Canyon Rd., Portland, OR 97221, USA

Summary

1. Parasites are thought to have numerous negative effects on their hosts. These negative effects may be associated with stress in a host.

2. We evaluated the effects of four species of flea ectoparasites (*Parapulex chephrenis*, *Synosternus cleopatrae*, *Xenopsylla conformis* and *Xenopsylla ramesis*) on non-specific responses of eight species of rodents (*Meriones crassus*, *Gerbillus dasyurus*, *Gerbillus andersoni*, *Gerbillus pyramidum*, *Gerbillus nanus*, *Acomys cahirinus*, *Acomys russatus* and *Mesocricetus auratus*) and measured faecal glucocorticoid metabolites concentrations (FGMC) produced by the hosts.

3. We found no effect of body mass of an individual rodent on FGMCs. Parasitism by fleas with a ‘stay on the host body’ exploitation strategy was associated with higher host FGMCs than parasitism by fleas that spent most of their life ‘off-host’. FGMCs among rodents infested by the same flea species were correlated positively with the phylogenetic distance of a given rodent from the principal host of this flea; changes in FGMCs were lower in the host species more closely related to the flea’s principal host. Changes in FGMCs of a host while parasitized were correlated with a host’s change in body mass, where hosts that lost more body mass had higher FGMCs.

4. Our results suggest that ectoparasitism can be stressful to their hosts. However, the occurrence of parasite-induced stress seems to depend on the identity of both host and parasite species and the evolutionary history of a host–parasite association. To our knowledge, this is the first multispecies study to evaluate the effect of ectoparasites on glucocorticoid hormones in small mammals.

Key-words: corticosterone, faecal glucocorticoid metabolites, fleas, parasitism, principal host, rodents, stress

Introduction

A parasite is a symbiont that, by definition, affects its host negatively (Poulin 2007). Indeed, the negative effect of

helminths and microparasites on host health, reproduction, density and mobility has often been demonstrated (Scott & Lewis 1987; Somczyński *et al.* 2006; Aguirre-Macedo *et al.* 2007; Råberg, Graham & Read 2009; Gooderham & Schulte-Hostedde 2011). In contrast, studies on ectoparasites provided contradictory results (see reviews in Degen 2006; Krasnov 2008; Robar, Murray & Burness 2011). A

*Correspondence author. E-mail: jstjuliana@sycamores.indstate.edu

negative effect of parasites on their hosts was evident in some but not other host–parasite associations, although ectoparasites undoubtedly incur negative effects on their hosts either directly by using energy and nutrients of hosts (Khokhlova *et al.* 2002) or indirectly by increasing activity of the immune system (De Lope, Møller & De la Cruz 1998), modifying behaviour (Christe, Richner & Oppliger 1996; Raveh *et al.* 2011) or decreasing food intake (Tripet, Glaser & Richner 2002).

Contradictions in the results of the studies of the negative effect of ectoparasites could result from, at least, three main reasons. First, different studies evaluated the effect of parasites using different host traits (e.g. Hawlena *et al.* 2006; Kam *et al.* 2010). However, a parasite might affect some but not other characteristics of the same host (Boughton, Atwell & Schoech 2006). Secondly, different hosts might be affected differently by the same parasite (Richner, Oppliger & Christe 1993 vs. Kędra *et al.* 1996 vs. Mazgajski *et al.* 1997). This does not mean that the parasite exhibits different exploitation strategies for each given host species, but rather suggests that some hosts cope with infestation better than others. Thirdly, different parasites may have different effects on the same host because of variation in optimal virulence among parasites, which in turn, might be determined by variation in physiological and/or behavioural traits of parasites (Khokhlova *et al.* 2004a). In addition, the reason behind differential effect of the same parasite on different hosts as well as differential effects of different parasites on the same host could be that a host–parasite association is often a product of a long history of parasite co-evolution with a host or group of hosts (Combes 2001). As a result, one of the adaptations of a given parasite to a given host can be that a specific parasite does not stimulate the host's anti-parasitic response (Ribeiro 1989; Beaty & Marquardt 1996). Likewise, a host may evolve defences and suffer less detrimental effects from its primary parasites. Alternatively, energetically costly defence effort may be of a little value if co-evolved parasites cope successfully with this defence, so a host often gives up its defences and surrenders to parasites (see review in Krasnov 2008).

In this study, we evaluated the effects of four species of flea ectoparasites on non-specific responses of eight species of rodent hosts and measured faecal glucocorticoid metabolites (FGM) produced by the hosts. Increased glucocorticoid hormone (i.e. 'stress' hormone) production by the adrenal gland is a key aspect of the adaptive biological stress response, yet it is often viewed and interpreted in a negative biological context (Carr & Summers 2002; Wingfield & Kitaysky 2002; Wielebnowski 2003). Stress in animals is known to result from any number of situations such as mating, predation, changes in social dynamics, raising young, competition, illness, lack of resources. (Wingfield *et al.* 1998; Wielebnowski 2003). Usually, when stress is experienced repeatedly in a very acute form, or chronically, does it lead to distress and subsequent nega-

tive biological and physiological consequences for the affected individual.

It has been hypothesized that the presence of parasites could represent a substantial and chronic stressor (e.g. Khokhlova *et al.* 2002; Boughton, Atwell & Schoech 2006; Brown & Fuller 2006; Hawlena *et al.* 2008). Glucocorticoid production is increased in association with stressful events due to increased activity of the hypothalamic–pituitary–adrenal axis (HPA), which causes their release from the adrenal cortex (Munck, Guyre & Holbrook 1984). Consequently, one way of assessing stress response is to measure the concentration of glucocorticosteroid hormones in an animal's blood or glucocorticoid metabolites in the urine or faeces (Harper & Austad 2000; Wasser *et al.* 2000; Koren *et al.* 2002; Wielebnowski & Watters 2007; Sheriff *et al.* 2011; Desantis *et al.* 2013).

We measured FGM concentrations in eight rodent species. We chose to use faecal hormone values, because the hormones extracted from faeces reflect stress experienced over duration of time, rather than just a short-term increase (Harper & Austad 2000; Millspaugh & Washburn 2004). Consequently, faecal metabolite concentrations could better reflect the chronic stress experienced by the rodents exposed to fleas. In addition, measurements of hormonal status via faeces are non-invasive, and the results of these measurements are unlikely confounded with a stress associated with handling of the animal. In our study, we used five gerbillines (*Meriones crassus*, *Gerbillus dasyurus*, *Gerbillus andersoni*, *Gerbillus pyramidum* and *Gerbillus nanus*), two murines (*Acomys cahirinus* and *Acomys russatus*), and a cricetine (*Mesocricetus auratus*) parasitized by each of four pulicid flea species (*Parapulex chephrenis*, *Synosternus cleopatrae*, *Xenopsylla conformis* and *Xenopsylla ramesis*). We hypothesized that rodents respond to flea parasitism by changes in hormonal status. The direct evidence for the effect of parasites on glucocorticoid production is limited (but see Barnard *et al.* (2002, 2003) and Brown & Fuller 2006 for examples with intestinal parasites). Consequently, we predicted that rodents infested by fleas would have higher FGM concentrations than uninfested rodents possibly due to integument damage (including physical discomfort), a reduction in energetic state and/or energy-costly immunological and behavioural defensive responses (Schmid-Hempel & Ebert 2003).

As noted above, the effect of a parasite on a host might be mediated by (i) host ability to cope with flea parasitism; (ii) pattern of host exploitation by a flea; and (iii) history of association between a flea and a host. We used host body mass and changes in host body mass while being infested with fleas as proxies for host ability to cope with parasitism. Indeed, smaller hosts are expected to manifest stronger and more efficient anti-ectoparasite grooming than larger hosts in order not to allow too many parasites per unit body surface both inter- and intraspecifically (Mooring *et al.* 2000, 2002; Mooring,

Blumstein & Stoner 2004). We predicted that larger species (across species) or individuals (within species) will have lower increases in FGM concentrations (FGMC) due to fewer fleas per surface area than smaller species or individuals. We also predicted that FGMC elevation in infested rodents will be associated with greater loss in body mass. This is because fleas that negatively affect host body mass are likely causing a decrease in the energetic state of their host (Khokhlova *et al.* 2002). Glucocorticoids are associated with increases in energy mobilization (Möstl & Palme 2002; Peckett, Wright & Riddell 2011), so that the energetic cost of fleas may actually be driving the hosts stress response.

Although adult fleas are obligatory haematophages, they usually alternate periods when they occur on a host's body and when they reside in its burrow/nest (Krasnov 2008). However, different flea species vary in the proportion of time they spend in each of these two places, so that 'body' and 'nest' flea species are distinguished (Ioff 1941; Krasnov, Khokhlova & Shenbrot 2004a). 'Body' fleas spend most (if not the entirety) of their life on a host, whereas the contact of a 'nest' flea with its host is intermittent, with it staying on a host only long enough to take a blood meal. Among flea species used in this study, *X. ramesis*, *X. conformis* and *S. cleopatrae* are 'nest' fleas, while *P. chephrenis* is a typical 'body' flea. Assuming that 'body' fleas cause more continuous irritation and extended grooming behaviour than 'nest' fleas, we predicted that rodents infested with the 'body' flea will have higher FGM concentrations than their counterparts that are infested with the 'nest' fleas.

A parasite species usually varies in its abundance and/or prevalence among different host species (Combes 2001). This variation has led to a distinction between (i) one host species in which the prevalence, intensity and/or abundance of a parasite are highest (the principal host) and (ii) all other hosts in which the prevalence, intensity and/or abundance of a parasite are lower (auxiliary hosts) (Dogiel, Petrushevski & Polyanski 1961; Krasnov *et al.* 2004b; Poulin 2005). Assuming that the ability of a parasite to stimulate immune response of a host depends on their history of host-parasite association (see above), we predicted that parasitism by a flea will result in lower FGM concentrations in their principal hosts as compared to their auxiliary hosts. For example, we should expect to find reduced 'unnecessary virulence' in co-evolved hosts and parasites (Ebert 2005). Moreover, it has been shown that flea abundance in an auxiliary host is negatively correlated with the phylogenetic distance between this host and the flea's principal host (Krasnov *et al.* 2004b) suggesting that the defences of the principal host against a flea are milder than those of an auxiliary host. Consequently, we also predicted that the level of FGM concentrations among rodents infested by the same flea species will correlate positively with the phylogenetic distance of a given rodent from the principal host of this flea. In other words, hosts

may be less stressed by parasites with which they have co-evolved.

Materials and methods

RODENTS AND FLEAS

We used rodents from our laboratory colonies. *Meriones crassus* (Sundevall, 1843) (32 males), *G. dasyurus* (Wagner, 1842) (eight females and 24 males), *G. nanus* (Blanford, 1875) (eight females and 24 males), *A. cahirinus* (Geoffroy 1803) (eight females and 24 males) and *A. russatus* (Wagner, 1840) (12 females and 20 males) were from laboratory colonies started in 1997–1999 and 2009 (see details in Krasnov *et al.* 2002). *Gerbillus andersoni* (Winton, 1902) (seven females and 25 males) and *G. pyramidum* (Geoffroy, 1803) (four females and 28 males) were captured in the wild in the Negev Desert and used in experiments at least a month after capture. All ectoparasites from wild-caught rodents were removed immediately after capture. Golden hamsters *M. auratus* (Waterhouse, 1893) (five females and 27 males) were available commercially.

Rodents in our colonies were maintained individually or in pairs in plastic cages (60 × 50 × 40 cm) at 25 °C with a photoperiod of 12 : 12 (L : D) h and with sawdust and dried grass as bedding material. They were offered millet seed and fresh alfalfa (*Medicago* sp.) *ad libitum* daily. No water was available as the alfalfa is known to supply sufficient moisture for their needs. *Acromys cahirinus* and *A. russatus* were offered commercial cat food once a week. Animals were maintained individually for at least 1 month prior to experiments.

All flea species used in this study are common parasites of rodent hosts in the central Negev Desert. *Parapulex chephrenis* (Rothschild, 1903) is a host specialist with the principal host being the Egyptian spiny mouse, *A. cahirinus*. This flea species is also found on a congeneric golden spiny mouse, *A. russatus*, and occasionally on the jird *M. crassus* and the gerbil *G. dasyurus* (Krasnov *et al.* 1997, 1999). Host-opportunistic *X. conformis* (Wagner 1903) and *X. ramesis* (Rothschild 1904) parasitize a variety of gerbilline species, but are the most abundant and prevalent on their principle host *M. crassus* (Krasnov *et al.* 1997, 1999). *Synosternus cleopatrae* (Rothschild, 1903) is restricted to sandy habitats and mainly exploits psammophilic hosts *G. andersoni* and *G. pyramidum*, with the former considered as the principal host of this flea (Krasnov *et al.* 1999).

All rodents used in experiments were adult (6–8 months old) and non-breeding. The majority of 256 rodents (187 individuals) were used in experiments only once, while 65 rodents were used twice and four rodents were used three times with at least 30 day-breaks between experiments. Rodents have never been exposed to fleas prior to experiments.

We used fleas from laboratory colonies started in 1999 (both *Xenopsylla* and *P. chephrenis*) and 2008 (*S. cleopatrae*) from field-collected specimens. *Xenopsylla ramesis* and *X. conformis* were collected from *M. crassus*, *Psammomys obesus* and *G. dasyurus*, and *P. chephrenis* were collected from *A. cahirinus* and *A. russatus*, while *S. cleopatrae* were collected from *G. andersoni* and *G. pyramidum*. In our colonies, fleas were maintained on their natural hosts (*X. ramesis* and *X. conformis* on *M. crassus* and *G. dasyurus*, *P. chephrenis* on *A. cahirinus* and *A. russatus*, and *S. cleopatrae* on *G. andersoni* and *G. pyramidum*) kept individually in plastic cages with a wire mesh floor over a pan with a

mixture of sand and dried bovine blood (larvae feeding medium) at 25 °C with a photoperiod of 12 : 12 (L : D) h. Every year, we added to colonies 100–150 fleas of each species collected in the wild from their natural hosts. This guaranteed genetic diversity of our flea colonies and likely prevented artificial selection for a single host species (see Arbiv *et al.* 2012). Every 2 weeks, all substrate and bedding material from the rodent's nest box and pan were collected into plastic boxes with perforated lids and transferred to an incubator (FOC225E; Velp Scientifica srl, Milano, Italy; 25 °C and 75% relative humidity) where the fleas developed. Once a week, newly emerged fleas were collected from these boxes and were used either for infestation of maintenance rodents or in experiments. Further details on breeding and maintenance of fleas are described elsewhere (e.g. Krasnov *et al.* 2002).

EXPERIMENTAL DESIGN

We divided rodents into 16 groups with two individuals of each rodent species in each group. Each group was exposed to one flea species, so that the infestation of each rodent by each flea species was replicated eight times over the course of 16 rounds. This totalled 32 rodent–flea combinations with eight individual rodents in each combination.

The cages used in the experiment were standard plastic rodent bins (60 × 50 × 40 cm) with barred metal tops. We placed a set of wire meshes elevated *c.* 0.5 cm above the floor of the cage with a sand layer of 1–2 mm. These wire meshes allowed the faeces to fall through. A plastic mesh screen (1 mm) was placed below the wire mesh to allow collection of the faeces and to avoid contamination with urine. This screen also allowed fleas free access to the rodent.

Each experimental test lasted 5 days. On day 1, a rodent was weighed (Precisa BJ 410C, Dietikon, Switzerland) to the nearest 0.1 g and transferred from its normal resident bin to the first experimental bin. After 24 h, we removed the rodent from the experimental bin and collected the control faeces (i.e. prior to flea exposure) from the bin. Then, we placed the rodent in the second (new) experimental bin with 50 fleas released into the sand. On day 4, we replaced the bottom screen (so we could collect a 24-h faecal sample) and released additional 25 fleas to compensate for fleas groomed off and killed by the rodent. This design guaranteed the uninterrupted flea exposure during 72 h. On day 5, we collected rodent faeces from the screen and returned the rodent to its original residence bin.

FAECAL COLLECTION AND PROCESSING

Faeces were collected from the bins no later than 24 h after defecation. Any faeces contaminated with urine was excluded from analysis. The faeces were placed in a tin foil bag and immediately frozen at –32 °C. After the last experimental round, all foil bags were soaked in liquid nitrogen for 30 s and then freeze-dried for 24 h. This procedure facilitated transportation and inhibited bacterial growth. We processed the samples at the endocrinology laboratory at the Chicago Zoological Society (Brookfield Zoo, Brookfield, IL, USA). After arrival at the laboratory, the faecal samples were stored in a –20 °C freezer. They were processed by first placing them on a piece of weigh paper. The weigh paper was then folded in half, and the faeces in the middle were pulverized with a rubber mallet. Then, the faeces were placed in a plastic test tube in a freezer for 5–8 days prior to actual weighing.

Before weighing, the faecal sample in the test tube was vortexed (this helped to further break up the sample). We then weighed out 0.5 g of the sample. We used the fine powder resulting from the hammering and vortexing. This helped to exclude any large chunks of alfalfa which might bias the results.

To extract steroid metabolites, 3 mL of 80% ethanol was added to the dry faecal material in polyethylene tubes. Tubes were placed on a rotator overnight and then centrifuged for 15 min at 500 g. One mL of supernatant was removed and diluted with 1 mL assay buffer [0.1 M phosphate-buffered saline (PBS) containing 1% BSA, pH 7.0].

A commercially available corticosterone EIA kit from 'Assay Designs' (Ann Arbor, MI, USA; catalogue # ADI-901-097) was used to analyse faecal samples for FGM concentrations. Manufacturer instructions and components were supplied and followed. Plates for the EIA were read on a photospectrometer plate reader (Dynex MRX Revelation Dynex Technologies, Chantilly, VA, USA) at 405 nm. The cross-reactivity of the corticosterone antibody is as follows: 100% corticosterone, 28.6% desoxycorticosterone, 1.7% progesterone, 0.28% tetrahydrocorticosterone, 0.18% aldosterone, 0.13% testosterone, 0.046% cortisol, and any other steroids were <0.03%.

This assay has been both biochemically and biologically validated. Assay sensitivity was 26.99 pg mL^{–1}, and the intra-assay coefficient of variation was 4.94% at 58.39% binding, 6.86% at 56.01% binding and 3.76% at 32.1% binding. The recovery of exogenous corticosterone (250–4000 pg mL^{–1}) was 74.91%. To biologically validate the assay, we conducted the adrenocorticotrophic hormone challenge on three of eight rodent species used in this study (J.R. St. Juliana, J.L. Bryant, N. Wielebnowski, B.P. Kotler, J.S. Brown, unpublished data). For *G. andersoni* (*n* = 4), we found a peak in FGM concentration after 9.25 h on average, with an average% increase from baseline of 368%. For *G. nanus* (*n* = 4), we found a peak in FGM concentrations after 6.25 h on average, with an average% increase from baseline of 308%. For *G. pyramidum* (*n* = 6), we found a peak in FGM concentrations after 9.83 h on average, with an average% increase from baseline of 374%. FGM concentrations are presented as nanogram per gram dry faeces.

DATA ANALYSES

In 11 of 256 rodents, FGMC prior to flea infestation was more than two SD greater than the average measurement for a given species. These were two *A. cahirinus* (pre-infestation FGMC were 2026.2 and 2915.2 vs. within-species mean FGMC ± SD = 617.5 ± 212.3), two *G. andersoni* (pre-infestation FGMC were 9016.1 and 10458.3 vs. within-species mean FGMC ± SD = 2724.3 ± 1856.3), one *G. dasyurus* (pre-infestation FGMC was 3571.5 vs. within-species mean FGMC ± SD = 869.4 ± 363.2), three *G. nanus* (pre-infestation FGMC were 3783.1, 3114.0 and 7675.9 vs. within-species mean FGMC ± SD = 1240.5 ± 639.4), two *G. pyramidum* (pre-infestation FGMC were 3090.6 and 3370.4 vs. within-species mean FGMC ± SD = 997.0 ± 338.6) and one *M. crassus* (pre-infestation FGMC was 1711.6 vs. within-species mean FGMC ± SD = 619.3 ± 199.2). These 11 individuals were excluded from the analysis.

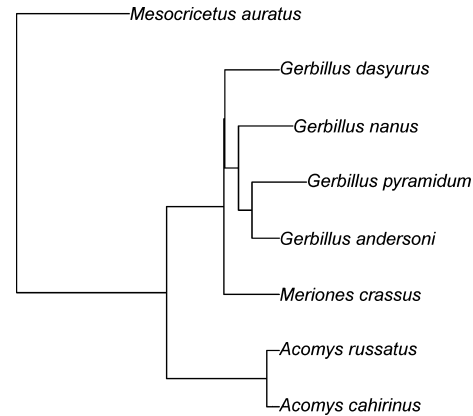
Prior to main analyses, we tested for sexual differences in FGMCs prior and after infestation by fleas as well as in proportional changes in FGMCs after infestations (see below) using linear mixed-effects models with rodent sex as a fixed explanatory variable and including rodent individual number as a random

effect in each model. These models were run for each rodent species separately. Although sexual difference in FGM concentrations might be expected (see Malisch *et al.* 2007), this was not the case in our study as well as in other studies on rodents (Dlugosz *et al.* 2012; Harris *et al.* 2012). No difference between male and female rodents in either FGM concentrations prior or after infestation by fleas or in its proportional change after infestation was found ($t = -1.47-0.30$, $t = -1.22-0.49$ and $t = -1.49-1.55$, respectively; $P >$ for all). Consequently, we pooled data for male and female rodents in subsequent analyses.

To test whether FGM concentrations change after exposure to fleas, we analysed FGMCs before and after exposure to four flea species using linear mixed-effects model with rodent species (eight species) and treatment (non-infested and infestation by one of the four flea species) as fixed explanatory variables and the individual rodent number as a random effect. We fitted the model using the lme function as implemented in 'nlme' package (version 3.1-106; Pinheiro *et al.* 2013) in the R software environment (version 2.14; R Development Core Team 2012). To test for differences between rodent or flea species as well as interaction between these factors in their effect on dependent variables, we used tests for multiple comparisons as implemented in 'PHIA' R package (version 0.1-5; Helios De Rosario-Martinez 2013). Prior to this analysis, pre- and post-infestation FGM concentrations values were log-transformed (Fig. 2 presents untransformed data for clarity).

Then, we calculated proportional changes in FGM concentrations and proportional changes in body mass after flea infestation as differences between post-infestation and pre-infestation values relative to pre-infestation values. To test for the effects of (i) 'body' vs. 'nest' fleas, (ii) initial body mass of an individual rodent, (iii) body mass change after infestation, and (iv) phylogenetic distance of a rodent species to the principal host of a flea (fixed effects) on proportional change in FGMC after exposure to fleas (dependent variable), we ran linear mixed-effects models including the individual rodent number as a random effect. We ran a set of models using the lme function as implemented in 'nlme' package as described above and searched for the best model using Akaike's information criterion (AIC). AIC (corrected for sample size) values and weights were calculated using respective functions of 'MuMIn' R package (version 1.9.5; Barton 2013). After visual examination of difference between rodent species in the relationship between change in FGMC and change in body mass after exposure to fleas, we further explored this relationship by running mixed-effects models with proportional change in FGMC as a dependent variable, proportional change in body mass as a fixed explanatory variable and the individual number of a rodent as a random effect. In addition, we explored within-flea species relationship between changes in FGMC and phylogenetic distance of a host from the principal host by running mixed-effects models with proportional change in FGMC as a dependent variable, phylogenetic distance from the principal host as a fixed explanatory variable and the individual number of a rodent as a random effect for each flea species separately.

We calculated phylogenetic distances between a principal host (*A. cahirinus* for *P. chephrenis*, *M. crassus* for *X. ramesis* and *X. conformis* and *G. andersoni* for *S. cleopatrae*) and each of the remaining seven host species. Phylogenetic distances between hosts were calculated from branch length of a phylogenetic tree using package 'ape' (Paradis, Claude & Strimmer 2004) implemented in R. The phylogenetic tree (topology and/or branch



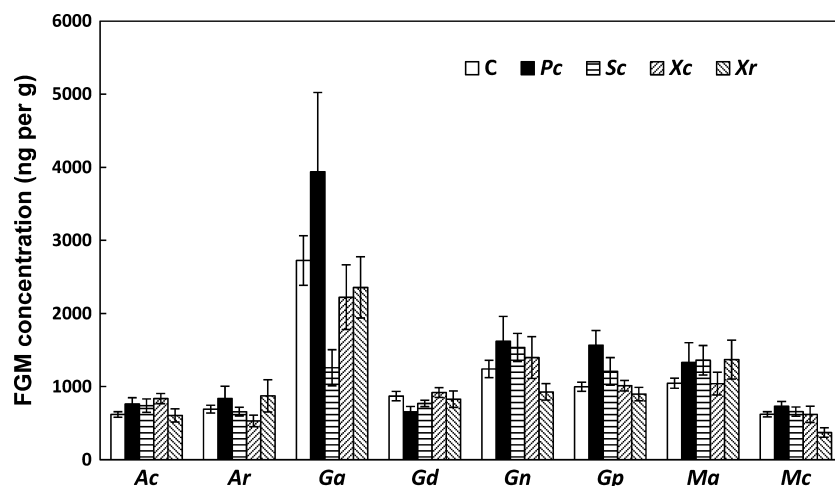


Fig. 2. Mean ($\pm$ SE) faecal glucocorticoid metabolites concentrations (FGMC) concentrations (ng per g faeces) in eight rodent species before (C) and after exposure to one of four flea species (*Pc* – *Parapulex chephrenis*, *Sc* – *Synosternus cleopatrae*, *Xc* – *Xenopsylla conformis* and *Xr* – *Xenopsylla ramesis*). Abbreviations of rodent species names are: *Ac* – *Acomys cahirinus*, *Ar* – *Acomys russatus*, *Ga* – *Gerbillus andersoni*, *Gd* – *Gerbillus dasyurus*, *Gn* – *Gerbillus nanus*, *Gp* – *Gerbillus pyramidum*, *Ma* – *Mesocricetus auratus* and *Mc* – *Meriones crassus*).

increased in comparison with either pre-exposure level or FGM concentration after exposure to any other flea species ($\chi^2 = 3.78$ – 12.21 , $P < 0.05$ for all; Fig. 2). In contrast, FGM concentrations after exposure to *S. cleopatrae* and both *Xenopsylla* species did not differ from pre-exposure values ($\chi^2 = 0.39$ – 1.48 , $P > 0.22$ for all) as well as between these fleas ($\chi^2 = 0.02$ – 2.31 , $P > 0.12$ for all; Fig. 2). Significance of the interaction between rodent and flea species arose because pre- and post-infestation FGM concentrations differed in some but not other rodents infested by some but not other flea species. For example, infestation by *P. chephrenis* caused elevation in FGM concentration as compared with pre-exposure level in *G. pyramidum* ($\chi^2 = 7.97$, $P < 0.01$; Fig. 2), but not in *A. russatus* ($\chi^2 = 0.53$, $P = 0.46$; Fig. 2). Another example is significant decrease in FGM concentration in *G. andersoni* after exposure to *S. cleopatrae* ($\chi^2 = 6.44$, $P < 0.01$; Fig. 2), but not to *X. conformis* or *X. ramesis* ($\chi^2 = 0.10$ – 0.51 , $P > 0.41$ for both; Fig. 2).

The best model explaining variation in changes in FGM concentrations before and after flea infestation contained the pattern of host exploitation ('body' vs. 'nest' flea), body mass change after flea infestation and phylogenetic distance of a rodent from the principal host of a flea (Table 2). Coefficient estimates of the model demonstrated that changes in FGMCs were higher under infestation by *P. chephrenis* than by other flea species (-0.19 ± 0.08 , $t = -2.1$, $P = 0.04$; 'body' flea was reference level), decreased with an increase in body mass change after infestation (-15.85 ± 5.25 , $t = -3.0$, $P = 0.004$) (Fig. 3) and increased with an increase of phylogenetic distance of a rodent species from the principal host of a flea (0.002 ± 0.0008 , $t = 2.4$, $P = 0.02$). Nevertheless, a decrease in FGMC with a decrease in body mass change after infestation was significant in three rodent species only (*A. russatus*, *M. crassus* and *M. auratus*; coefficient estimates were -33.89 ± 13.93 , -23.16 ± 10.49 and -37.38 ± 11.12 , respectively; $t = -2.43$ to -2.10 , $P < 0.04$ for all; Fig. 3). Similarly and despite overall significant relationship between across-rodent proportional

Table 2. Results of search for the best linear mixed-effects model of the effect of flea strategy of host exploitation ('body' vs. 'nest'; BN), initial rodent body mass (BM), rodent body mass change after flea infestation (BMC) and phylogenetic distance of a rodent from the principal host of a flea (PD) on proportional change in faecal glucocorticoid metabolites concentrations after infestation by fleas

Model	d.f.	AICc	AICw	Δ AIC	LR χ^2	P
BN + BMC + PD	6	444.27	0.50	0.00	22.04	0.001
BN + BM	7	446.38	0.17	2.11	22.05	0.002
+ BMC + PD						
BMC + PD	5	446.65	0.15	2.38	17.56	0.003
BN + BMC	5	448.45	0.06	4.18	15.76	0.007
BM + BMC + PD	6	448.74	0.05	4.47	17.57	0.007
BN + BM + BMC	6	450.18	0.03	5.91	16.13	0.01

AICc, Akaike Information Criterion corrected for sample size; AICw, Akaike Information Criterion weight; LR χ^2 – log-likelihood ratio χ^2 . Likelihood ratio tests are against the null model (intercept only). Listed are models with Δ AIC < 6.

changes in FGMC and 'phylogenetic distance of a rodent from the principal host of a flea (Fig. 4), when considered separately for each flea species, this relationship appeared to be significant only in *X. ramesis* (coefficient estimate = 0.002 ± 0.001 , $t = 2.15$, $P = 0.04$), but not in *P. chephrenis*, *S. cleopatrae* and *X. conformis*.

Discussion

The results of this study partly supported some of our predictions. We found that, in general, flea infestation affected FGM concentrations of individual rodents in some but not other host–flea associations. The occurrence and magnitude of this effect depended on the identity of both host and flea species. In addition, FGM concentrations in a host infested by a given flea species were correlated with the phylogenetic distance between the host and the flea's principal host. Furthermore, rodents with greater reduction in body mass under flea infestation

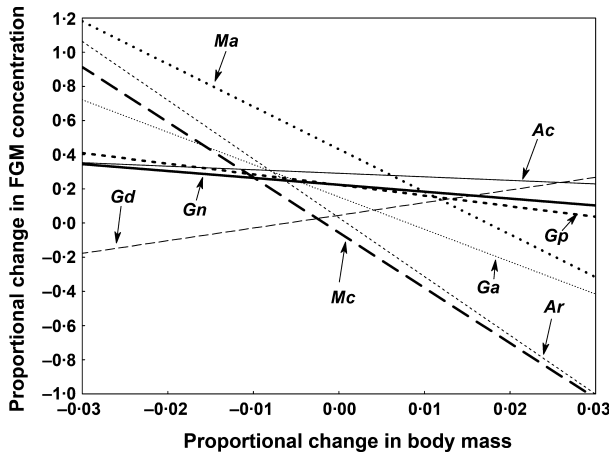


Fig. 3. Relationships between proportional change in faecal glucocorticoid metabolites concentrations (FGMC) and proportional change in body mass after infestation by fleas. For clarity, data points are not shown. See Fig. 2 for abbreviations of rodent species names.

showed greater changes in FGMCs. The effect of flea infestation on rodent FGM concentrations differed between flea species with different patterns of host exploitation, tending to be stronger for the 'body' flea (*P. chephrenis*) than the 'nest' fleas (*S. cleopatrae* and both *Xenopsylla*). Our predictions about association of body mass with FGMC changes under flea infestation either inter- or intraspecifically were not supported by our findings. Below, we will further discuss the physiological, ecological and evolutionary reasons for the observed patterns.

In general, there was a large degree of individual variation in FGM concentrations among rodents of the same species, both before and after infestation by either flea species. Such high levels of individual variation in adrenal responses and glucocorticoid secretion have been well-documented in a variety of species, and it is known that

glucocorticoid production can be influenced by a wide variety of factors (Palme 2005; Touma & Palme 2005). Therefore, FGM concentrations of any individual rodent may be lower or higher on any given day due to several uncontrollable effects. For example, simple handling a called for by the experimental protocol (e.g. transfer to a new cage) might be a stressful factor for some, but not other individuals, in accordance with their particular temperament or personality type (e.g. Bohus *et al.* 1987). A study on rats found that production of glucocorticoids systematically varied between individuals with different degrees of neophobia (Cavigelli & McClintock 2003). Similarly, differences in FGM concentrations across flea-infested individuals might occur due to several physiological, biological and behavioural reasons. In fact, it has been established that chronically elevated stress hormone levels (i.e. glucocorticoid levels) have been linked to an array of problems in animals, including decreased health (Wielebnowski 2003) and impaired reproduction (Charbonnel *et al.* 2008; Boonstra 2013).

The effects of parasites (including fleas) on their host can be direct, such as using energy and nutrients of hosts (Khokhlova *et al.* 2002), and indirect, such as increasing activity of the immune system (Wedekind 1992; Lochmiller & Deerenberg 2000) and decreasing food intake due to distraction and allocation of time to antiparasitic behaviour (Raveh *et al.* 2011). Infestation by haematophagous parasites may entail energy loss in a host via (i) loss of the energy of blood extracted by a flea, (ii) loss of energy spent for the haemopoietic compensation for the lost blood, and (iii) loss of the energy for behavioural and/or immune responses (including cutaneous immune response). If increasing glucocorticoids help an animal to mobilize energy (see Solomon *et al.* 2011), then some elevation in FGM concentrations under flea parasitism can be expected. However, our current results do not allow us

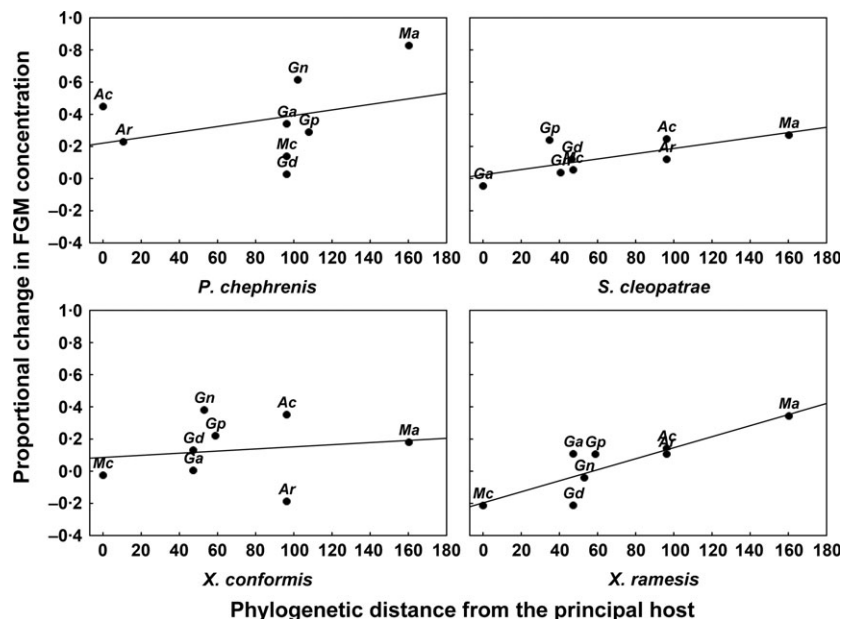


Fig. 4. Relationship between proportional change in faecal glucocorticoid metabolites concentrations (FGMC) in rodents belonging to eight species after exposure to *Parapulex chephrenis*, *Synosternus cleopatrae*, *Xenopsylla conformis* or *Xenopsylla ramesis* and phylogenetic distance of these rodents from the principal hosts of fleas (*Acomys cahirinus* for *P. chephrenis*, *Gerbillus andersoni* for *Synosternus cleopatrae* and *Meriones crassus* for *Xenopsylla conformis* and *Xenopsylla ramesis*). See Fig. 2 for abbreviations of rodent species names.

to draw any conclusions about a direct association between energy cost of parasitism and the level of glucocorticoid secretion. In an earlier study, Khokhlova *et al.* (2002) demonstrated that energy requirements of *G. dasyurus* parasitized by *X. ramesis* were higher than those of non-parasitized gerbils. In contrast, Hawlena *et al.* (2006) did not find any energy-related effect of parasitism by *S. cleopatrae* on *G. andersoni*. In our study, *G. dasyurus* did not show any significant change in total FGM concentrations after infestation by *X. ramesis*, while FGM concentration in *G. andersoni* decreased after exposure to *S. cleopatrae* (Fig. 2). Nevertheless, we found a positive relationship between body mass loss and changes in FGM concentrations after exposure to fleas at least in some rodent species. This finding therefore does suggest a link between the effect of fleas on glucocorticoid production and other energy-related physiological responses.

Physiological mechanisms behind changes in FGM concentrations in rodents exposed to flea parasitism could be related to complex relationships between glucocorticoid production, stress and skin immune response, and a trade-off between the benefits and costs of elevated glucocorticoids (Sapolsky, Romero & Munck 2000; Weil, Martin & Nelson 2006). In general, glucocorticoids tend to suppress inflammation (Barnes 1998; Tracey 2002). Acute stressors may trigger short-term elevation of glucocorticoids, which, in turn, can boost some aspects of immunity due to changes in the distribution of leucocytes (Dhabhar & McEwen 1997; Dhabhar 1998). In contrast, chronic stressors and prolonged high glucocorticoid concentrations lead to suppression of skin immune function (Dhabhar & McEwen 1999, 2001) and immune function in general (Boonstra 2013). These mechanisms seem to be especially important for immune response against haematophagous ectoparasites because host–ectoparasite immune interactions occur primarily at the cutaneous interface. Thus, on the one hand, glucocorticoids may limit parasite-induced inflammation and skin damage and thus play a role in development of host tolerance to parasites (Baum & de Roode 2011), while on the other hand, elevated glucocorticoid concentrations may facilitate survival, feeding and/or reproduction of parasites due to their immunosuppressive effect. In addition, if a parasite attack itself is stressful for a host, it may result in increased glucocorticoid production (Weil, Martin & Nelson 2006; Brown & Fuller 2006; but see Monello *et al.* 2010). Variation in the patterns and changes of FGM concentrations after exposure to parasites among rodents in this study may be related to the bidirectional effect of glucocorticoids produced as a result of acute vs. chronic stress and the respective effect on immune function. In addition, this effect can be mediated by (i) a pattern of natural parasite pressure in a rodent, (ii) a pattern of the response of a particular host to an attack by a particular flea, and (iii) a pattern of host exploitation by a flea.

The number of parasites that a host species must deal with may be related to the stress parasites cause their

host. Diversity of parasite challenges (e.g. parasite species richness) is positively correlated with genetic diversity of the major histocompatibility complex (MHC) (Wegner *et al.* 2003; Šimková, Ottová & Morand 2006; Goüy de Bellocq, Charbonnel & Morand 2008). High parasite diversity can also be associated with a strong investment in immunity (Morand & Poulin 2000; Møller & Rózsa 2005; Šimková *et al.* 2008). In other words, a new parasite can be a strong stress-factor for a ‘parasite-poor’ host, but less so for a ‘parasite-rich’ host. Among gerbils used in this study, *G. andersoni*, *G. pyramidum* and *G. nanus* are mainly parasitized by one or two flea species (Krasnov *et al.* 1999; Hawlena, Abramsky & Krasnov 2005; Kam *et al.* 2010), while at least seven flea species were recorded on *G. dasyurus* (Krasnov *et al.* 1999; Khokhlova *et al.* 2004b). If *G. dasyurus* is indeed selected to cope with multiple flea challenges, while the three other gerbils are not, then this hypothesis may at least partly explain why FGM concentrations increased after exposure to at least one flea species in *G. andersoni*, *G. pyramidum* and *G. nanus*, but not in *G. dasyurus*. We recognize, however, that this explanation is highly speculative and requires further experimental support. In addition, it is unclear why both ‘parasite-poor’ spiny mice did not demonstrate changes in post-infestation FGM concentrations similar to those observed in ‘parasite-poor’ gerbils.

Differential patterns in the changes observed in FGM concentrations after flea infestation across rodents and flea ‘treatments’ might be associated with the history of the relationship between a given rodent and a given flea. A long history of adaptation of a parasite to its principal host may result in its ability not to stimulate immune response in this host (Ribeiro 1987; Khokhlova *et al.* 2008), and consequently, not to be a stressor. This reduced response to fleas could result from reduced virulence of the fleas, reduced immune response by the host or increased host defences that make the parasitism less stressful. Reduced response to fleas can also occur in host species that are phylogenetically close to the principal host because phylogenetic relatedness among ‘species is generally a good reflection of their ecological and physiological similarity (Harvey & Pagel 1991). Consequently, infestation by a flea is expected to be more stressful for hosts phylogenetically more distant from the principal host of this flea. Indirect evidence for this is that the abundance of fleas in a host has been found to decrease with the distance of this host from the principal host of this flea (Krasnov *et al.* 2004b). One of the mechanisms behind this may be enhancement of a cell-mediated anti-flea response in distant relatives of the principal host due to acute stress caused by flea infestation (Dhabhar & McEwen 1997).

The most likely reason behind a stronger effect of *P. chephrenis* on FGM concentration of its hosts in comparison to other flea species is its strategy of host exploitation as a ‘body’ flea (see above). The ‘body’ strategy allows fleas to avoid prolonged searching and/or waiting

for a host, but it exposes them to the host's behavioural defences such as anti-parasitic grooming (Hart 1990). To withstand host grooming, 'body' fleas have evolved morphological tools such as sclerotized combs (ctenidia) to anchor themselves within host hair (Traub 1972). Although *P. chephrenis* does not possess ctenidia (perhaps, due to phylogenetic constraints because ctenidia are not characteristic for the entire *Pulicidae* family), its entire body, in contrast to other fleas used in this study, is covered with dense and long rigid bristles that likely play an anchoring role. As a result, hosts infested with *P. chephrenis* are forced to groom excessively, although excessive grooming may not be effective enough against a flea with strong anchoring ability (Nikitina & Nikolaeva 1981). Numerous, prolonged and irritating bites of a 'body' flea and subsequent inflammation may induce increased glucocorticoid production. In addition, excessive grooming may be costly to the host in other ways, for example, due to loss of water via saliva (Ritter & Epstein 1974) and energy expenses (Giorgi *et al.* 2001), so that excessive grooming can be stressor in itself. Our study does not provide us with any indication that glucocorticoids themselves may be causal in influencing animal behaviour. Furthermore, we recognize that generalization about the difference in the effects of the 'nest' vs. 'body' fleas on host hormonal status found in our experiments should be done cautiously because we had at our disposal only a single 'body' flea species.

In conclusion, our results suggest that the level of stress in a rodent host as a response to flea infestation is variable, both across species and across individuals. This variation may be explained in part by (i) differences in host responses to flea infestation among rodent-flea associations (evolutionary history), (ii) differences in the responses to a 'non-familiar' flea among rodents and (iii) differences in the patterns of host exploitation among fleas. In addition, the response may be related to the energetic costs of fleas.

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